AI-Based Pattern Recognition and Algebraic Limits of Frequency Analysis

What Is Learnable — and What Is Not

Johann Pascher
ORCID 0009-0000-6518-4064

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Status markers

[B] *Proved* — algebraically proved.
[K] *Confirmed* — numerically verified.
[S] *Speculative* — open.
[E] *Established* — textbook or consensus knowledge.
[L] *Literature reference* — external primary evidence.
[Q] *Source* — external primary value.

Abstract

AI-based pattern recognition in frequency analysis runs into the same algebraic limits as classical methods — but for a different reason and without knowing it. This document states the three limits precisely, supports them with current literature, and sketches what a Galois-informed AI architecture could achieve.

Results:

1. Theorem A **[B]**: The standard HRV frequency bands (Task Force 1996) have band limits with exclusively 5-limit prime factors $\{2, 3, 5\}$ and thus lie in the Galois grid of Doc. 358 Theorem B. They are compatible with the grid, but not determined by it: rounded decimal values almost always lie in the 5-limit (Doc. 360).
2. Theorem B **[B]**: The resolution limit $100\xi \approx 1.33\,\%$ (Doc. 343) has the Farey height $Q_c \approx 11.1$. A classification with this tolerance requires a frequency resolution $\Delta f/f \leq 100\xi$.
3. Theorem C **[B]**: Group-equivariant networks (G-CNNs) for continuous groups (E(3), SO(3)) demonstrably reduce the number of parameters and improve generalisation. For finite Galois groups GF(3^k) this architecture does not yet exist — this is the identified gap.

4. Theorem D **[B]**: Whether the admissibility criterion of Doc. 358 Theorem B is learnable is decided by the hypothesis class, not by the amount of data. Without a structural assumption the PAC bound requires about 84 000 labelled examples; with the structural assumption "prime factors from a fixed set" at most 407 suffice, and from positive examples alone the prime set $\{2, 3, 5, 7, 11, 13\}$ is reliably recovered after 30 examples.
5. Architecture sketch **[S]**: A Galois-informed neural network uses GF(3^k) orbits as a hard constraint in the input layer. This reduces the parameter space and makes the admissibility criterion not learned but enforced.
   Verification script: `pruef_359_ki_grenzen.py` (18/18).

1 Starting point: what AI does today

Physics-Informed Neural Networks (PINNs)

PINNs use known differential equations as a hard constraint in the loss function **[L]** [Raissi2024]. They know nothing of algebraic group structure. A PINN can incorporate an oscillation equation, but not the condition "prime factors only in $\{2, 3, 5, 7, 11, 13\}$".

Group-equivariant networks (G-CNNs)

G-CNNs build symmetry groups directly into the architecture, so that the output is equivariant under group operations **[L]** [Finzi2022]. For continuous groups such as E(3) (rotation, translation) and SO(3), such architectures have been established since 2021 and show measurable improvements in molecular modelling and materials science.

For *finite* fields GF(3^k), a comparable architecture is entirely missing. This is the gap that this document identifies.

HRV analysis with AI

Multi-resolution HRV analysis with machine learning shows that different resolution levels yield different features **[L]** [Gupta2024]. Current AI-based HRV systems use the Task Force bands VLF/LF/HF as a fixed input quantity, without knowing why these limits lie where they do **[L]** [Villanueva2026].

2 Theorem A: HRV band limits in the Galois grid

Theorem A. [B] The standard HRV frequency bands **[L]** [TaskForce1996] have the following limits as simplest rational approximations:

$$f_{\text{VLF,max}} = \tfrac{1}{25}\,\text{Hz} = 0.04\,\text{Hz} \qquad \text{prime factors: } \{5\} \tag{1}$$

$$f_{\text{LF,max}} = \tfrac{3}{20}\,\text{Hz} = 0.15\,\text{Hz} \qquad \text{prime factors: } \{2, 3, 5\} \tag{2}$$

$$f_{\text{HF,max}} = \tfrac{2}{5}\,\text{Hz} = 0.40\,\text{Hz} \qquad \text{prime factors: } \{2, 5\} \tag{3}$$

All prime factors lie in $\{2, 3, 5\}$ — the 5-limit grid, and at the same time in the Galois grid $\{2, 3, 5, 7, 11, 13\}$ of Doc. 358 Theorem B. The band ratios $f_{\text{LF}}/f_{\text{VLF}} = 15/4$ and $f_{\text{HF}}/f_{\text{LF}} = 8/3$ likewise lie in the 5-limit.

Proof. Direct prime factorisation (verification script A1–A5).

Consequence for AI: Current HRV AI models use these band limits as free hyperparameters, learned from data or set manually **[L]** [Villanueva2026]. A Galois-informed network could adopt them as fixed constants. This is a setting, not an algebraic necessity: 8 of the 9 numbers with one significant digit and a third of all two-decimal values lie in the 5-limit, and in rad/s the limits are irrational (Doc. 360). As an "inductive bias" in the sense of learning theory **[L]** [ILCRBlog2023] the adoption is admissible, but not algebraically grounded.

3 Theorem B: Resolution limit and Farey height

Theorem B. [B] The FFGFT resolution limit $100\xi \approx 1.33\,\%$ (Doc. 343 Theorem G′) has Farey denominator height $Q_c = \pi/\sqrt{6 \cdot 100\xi} \approx 11.1$. Frequency ratios with denominators $> Q_c$ cannot be separated in the Galois grid. The HRV band limits have denominators 5, 20, 25 — all in the separable range. A classification with the tolerance 100ξ is meaningful only if the frequency resolution satisfies $\Delta f/f \leq 100\xi$.

Practical consequence [K]: With $\Delta f = 1/T_w$ for the window length T_w (in Welch's method the segment length) this gives $T_w \geq 1/(100\xi f)$, i.e. 12.5 min at 0.10 Hz and 15 min at 0.083 Hz (Doc. 360). Shorter windows cannot distinguish grid elements. Whether the threshold appears empirically in AI-based HRV analyses is not established **[S]**.

4 Theorem C: What group-equivariant networks achieve — and where the gap is

Theorem C. [B] A G-equivariant network with group G has, in a fully connected layer $\mathbb{R}^{|G|} \to \mathbb{R}^{|G|}$, only $|G|$ free parameters instead of $|G|^2$, because the weight matrix is completely determined by the group structure **[L]** [Finzi2022]. For $G = (\mathbb{Z}/13)^*$ with $|G| = 12$ this means a factor 12 parameter reduction compared with an ungrouped network.

Proof. The group $(\mathbb{Z}/13)^*$ is cyclic of order 12 with generators $\{2, 6, 7, 11\}$ (verification script C2). A G-equivariant network has a weight matrix of the form $(W_{ij}) = w_{j-i \bmod 12}$ — a circulant network with $|G| = 12$ parameters (verification script C3).

What exists for E(3) and SO(3) [L] [Finzi2022]: Equivariant networks for continuous groups (rotations, translations) are standard in molecular modelling and materials science. They use spherical harmonics and Clebsch–Gordan products as building blocks.

What is missing: An analogous architecture for $\mathrm{GF}(3^k)^*$ — a finite, non-continuous group. The challenge: $\mathrm{GF}(3^k)^*$ has no natural embedding into a Euclidean space on which standard AI layers operate. The orbits of the Frobenius action ($x \mapsto x^3$) must be built into the architecture as discrete symmetries, not as continuous transformations.

Architecture sketch [S]: (Section 6)

5 Theorem D: Learnability depends on the hypothesis class

Theorem D. [B] On the bounded grid of reduced fractions p/q with $p, q \leq 100$, the admissibility criterion of Doc. 358 Theorem B (prime factors only in $\{2, 3, 5, 7, 11, 13\}$) is learnable. How many examples this requires is decided by the hypothesis class, that is, by the prior knowledge a learning system brings with it, and not by the amount of data alone.

Justification. The grid comprises 6087 reduced fractions, of which 1453 are admissible (verification scripts D1, D3). One hundred examples cover less than $7\,\%$ of the admissible ratios (D2). For a finite hypothesis class $\mathscr{H}$ in the realisable case, $m \geq (\ln|\mathscr{H}| + \ln(1/\delta))/\varepsilon$ labelled examples suffice to reach an error below ε with probability $1-\delta$ **[L]** [ShalevShwartz2014]. With $\varepsilon = \delta = 5\,\%$ this gives two very different cases.

Without a structural assumption, i.e. when any subset of the 6087 fractions may be the criterion, $|\mathscr{H}| = 2^{6087}$ and the VC dimension equals 6087. The bound requires about 84 000 labelled examples (D4), and from positive examples alone the criterion cannot be obtained at all, because every unseen ratio could equally well be admissible. This is the situation of a generic network without built-in prior knowledge.

With the structural assumption "admissible are exactly the fractions whose prime factors lie in a fixed set P", where P is a subset of the 25 primes up to 100, one has $|\mathscr{H}| = 2^{25}$, and at most 407 examples suffice (D5). In this class even a procedure that sees only positive examples learns: it takes as P the smallest prime set that explains all examples seen. For randomly drawn admissible ratios it recovers $\{2, 3, 5, 7, 11, 13\}$ after 10 examples in $94.6\,\%$ and after 30 examples in all of 1000 runs (D6). Such a system excludes $17/11$ without ever having seen a negative example, because the 17 occurs in no example.

Gold's theorem **[L]** [Gold1967] concerns identification of a language in the limit from infinite sequences of positive examples; it applies only when the hypothesis class is unbounded and yields no non-learnability for the bounded grid.

What this means: The knowledge that only the primes up to 13 occur does not reside in the amount of training data but in the form of the hypotheses. A network lacking this form needs more than two orders of magnitude more labelled examples, and with measurement noise above the resolution limit of Theorem B it does not recover the missing information at all.

Consequence: The admissibility criterion belongs in the architecture as a structural constraint, not as subsequent regularisation. Pointwise regularisation (ℓ_1, ℓ_2) favours small weights but no prime set, and therefore enforces no algebraic inadmissibility.

6 Architecture sketch: Galois-informed neural network

The following sketch is speculative **[S]** and is meant to indicate the direction, not to describe a finished architecture.

Input layer: Galois projection

The input layer maps measured frequency ratios f_i/f_j onto the nearest rational number in the Galois grid:

$$\pi_G(x) = \operatorname*{arg\,min}_{p/q \in \mathscr{R}_G} |x - p/q|, \quad \mathscr{R}_G = \{p/q : \text{prime factors} \subseteq \{2, 3, 5, 7, 11, 13\}\}$$

This is not a learning operation — it is an algebraic preprocessing step. Ratios outside the grid are projected onto their nearest grid approximation and tagged with a distance flag.

Middle layers: $(\mathbb{Z}/13)^*$ equivariance

The middle layers are $(\mathbb{Z}/13)^*$-equivariant: circulant matrices with 12 instead of 144 parameters per layer. This corresponds exactly to the principle of G-CNNs for the group $G = (\mathbb{Z}/13)^*$. The Frobenius action $x \mapsto x^3$ is the natural symmetry (Doc. 336, 341) and is built in as weight sharing.

Resolution filter: 100ξ threshold

Before the Galois projection: a low-pass filter with cutoff $\Delta f/f = 100\xi \approx 1.33\,\%$. Everything below this threshold is treated as embedding noise and is not classified. The threshold is a setting of FFGFT; an empirical counterpart in HRV data is not established (Doc. 360).

Stability assessment: Arnold tongue width

The output of the network contains not only a classification but also a stability assessment: the algebraically computed Arnold tongue width (Doc. 328) of the classified ratio. This immediately gives the user an uncertainty estimate — not learned by the network, but computed from the algebra.

What this achieves compared with standard AI

	Standard AI	Galois-informed
Admissibility criterion	learned (incomplete)	enforced (complete)
Resolution limit	not provided	set algebraically
Stability prediction	only for cases seen	for all cases (Arnold tongue)
Number of parameters	$\lvert G\rvert^2 = 144$	$\lvert G\rvert = 12$
Interpretability	limited	complete

7 Application: HRV

For heart rate variability analysis, concrete consequences follow:

1. **Band limits as fixed constants:** VLF/LF/HF limits are not learned from data (Theorem A); they are compatible with the grid, but a setting. The search space of the network shrinks accordingly.
2. **Required resolution:** $\Delta f/f \leq 1.33\,\%$ (Theorem B), i.e. ≈ 0.0013 Hz at 0.10 Hz and a window length of at least 12.5 min.
3. **New biomarkers:** Instead of the LF/HF ratio (a single number), a Galois-informed network could output the *orbit type* of the dominant frequency ratio — an algebraic biomarker that indicates in which Galois layer ($k = 3, 4, 5, 6$) the heart is currently oscillating. This is new and has no counterpart in the literature **[S]**.
4. **Stability prediction [S]:** Whether a current HRV pattern is robust (wide Arnold tongue) or fragile (narrow tongue). The premise that HRV frequency ratios lock onto the grid is untested; the data in Doc. 360 do not support it, and a prospective test is described there.

8 Scope

- This document implements no AI architecture and makes no clinical claims.
- The architecture sketch in §5 is speculative **[S]** — it outlines a direction, not a finished system.
- The algebraic orbit biomarker in §6 is a hypothesis; its physiological significance is open. The exploratory test on HRV data (Doc. 360) gave a negative result.

- Theorem D compares sample bounds on the bounded grid $p, q \leq 100$; over an unbounded grid with arbitrarily large primes the hypothesis class is infinite and the statements must be re-examined.

9 Summary

Theorem	Statement	Status
A	HRV band limits in the 5-limit grid: compatible, not determined	**[B]**
B	Resolution limit $100\xi \approx 1.33\,\%$, $Q_c \approx 11.1$; classification requires $\Delta f/f \leq 100\xi$	**[B]**
C	G-CNNs for GF(3^k): parameter reduction 12x; architecture missing	**[B]/[S]**
D	Learnability of the admissibility criterion: ~84 000 examples without structural assumption, ≤407 with prime-factor structure	**[B]**
Arch.	Galois-informed network: sketch	**[S]**

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Verification scripts

`2/python/Dok359_Skripte/pruef_359_ki_grenzen.py` — 18 assertions: A1–A5 (HRV grid), B1–B4 (resolution), C1–C3 (group equivariance), D1–D6 (grid, coverage, PAC bounds, learning from positive examples).

Note on the use of AI assistance

AI assistance (Claude, Anthropic) was used in preparing this document for wording, LaTeX typesetting, literature research and the creation of the verification script. The substantive claims, the status assignments and the responsibility for the text lie with the author.